Laboratory Assignment 4

Completed by Elvira Oliinyk

4 INCIDENT MANAGEMENT IN INFORMATION AND COMMUNICATION SYSTEMS USING A SIEM SYSTEM

4.1 Objective

To learn how to detect cybersecurity incidents using SIEM systems.

4.2 Laboratory Procedure

4.2.1 Initial configuration

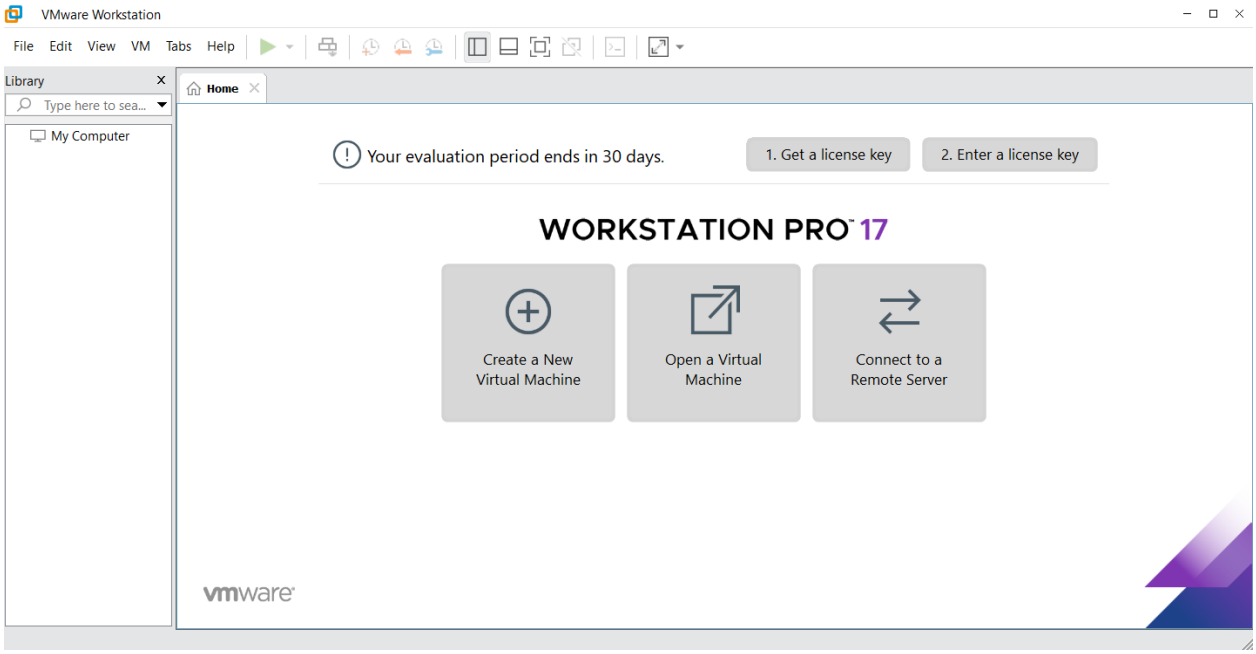


Figure 4.1 – Installing VMware Workstation

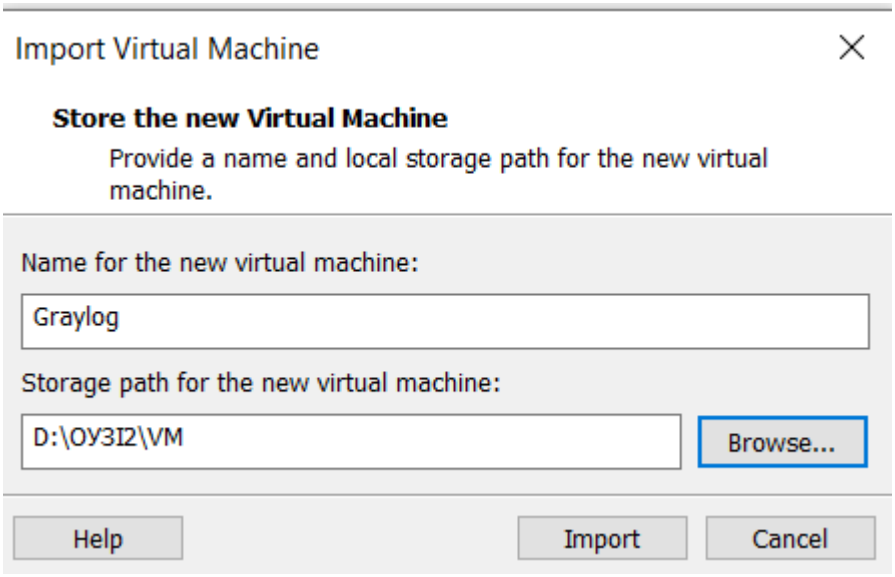


Figure 4.2 – Creating the “Graylog” virtual machine

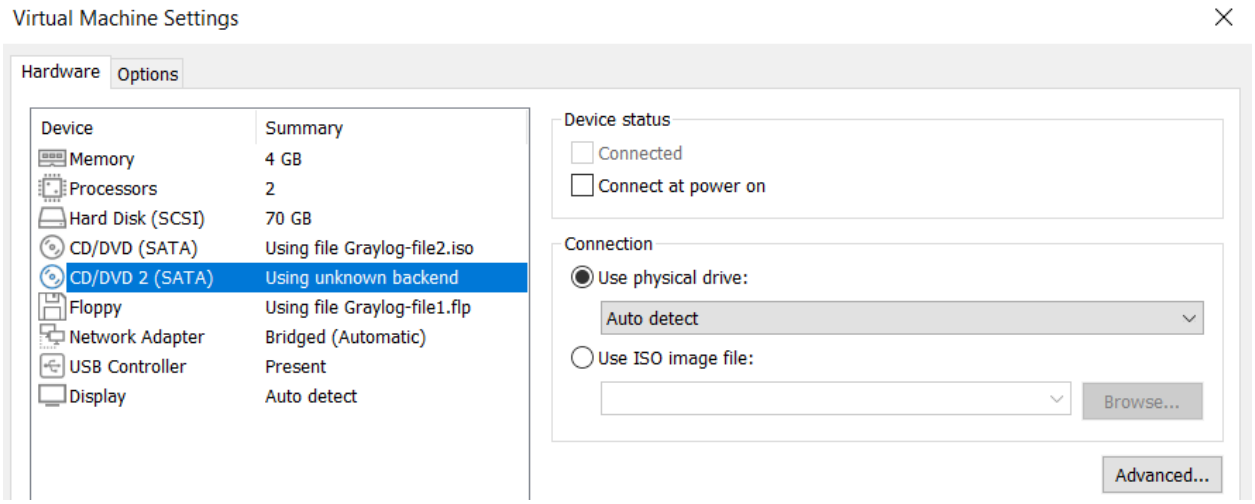
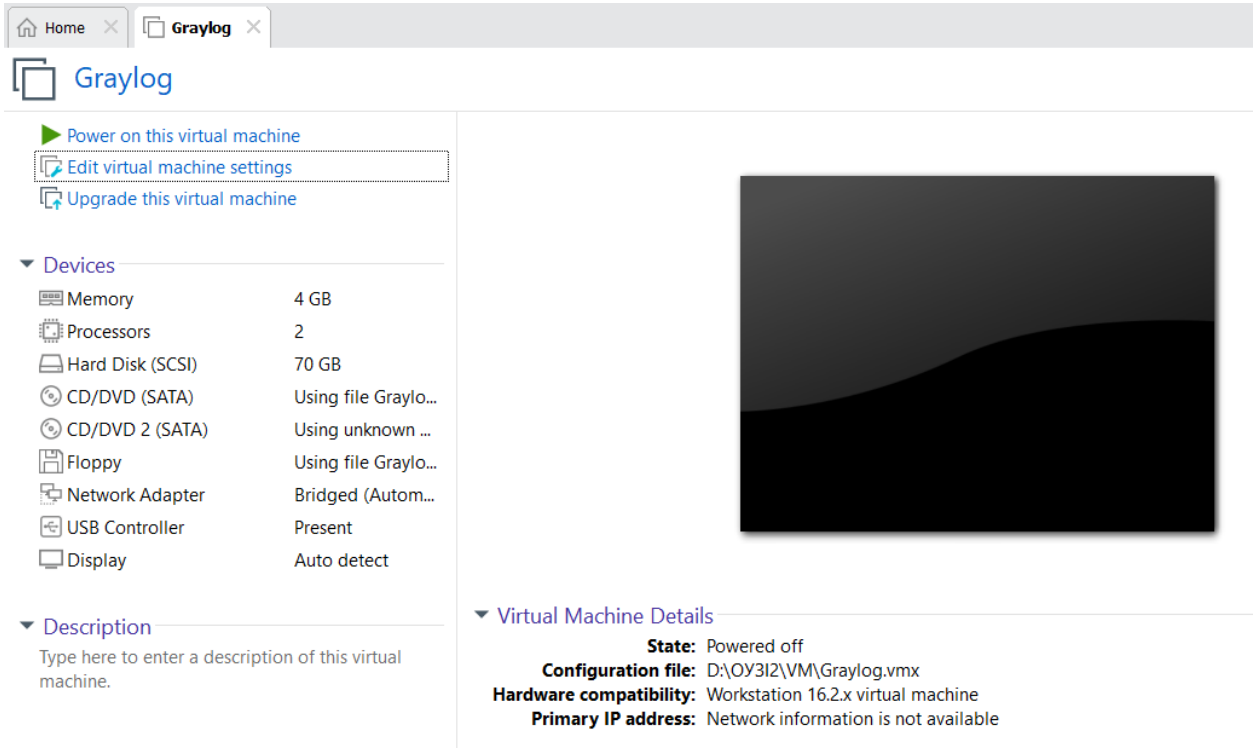


Figure 4.3 – Configuring the virtual machine



4.4 – Result of installing the “Graylog” virtual machine

Before starting the practical work, VMware Workstation was installed and the “Graylog” virtual machine was created using this virtualization software.

4.2.2 Configuring the virtual machine

```
Ethernet adapter VMware Network Adapter VMnet1:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::3fc9:1b05:ebc0:8484%52
   IPv4 Address. . . . . . . . . . . : 192.168.146.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
```

Figure 4.5 – IP address of the host machine

As shown in the example, the IP address of the Windows PC is 192.168.146.1/24. An available address from this subnet was selected for the server: 192.168.146.177.

```
C:\Users\Dell>ping 192.168.146.177

Pinging 192.168.146.177 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.146.177:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\Dell>
```

Figure 4.6 – Verifying that the selected IP address is not in use by another device on the network

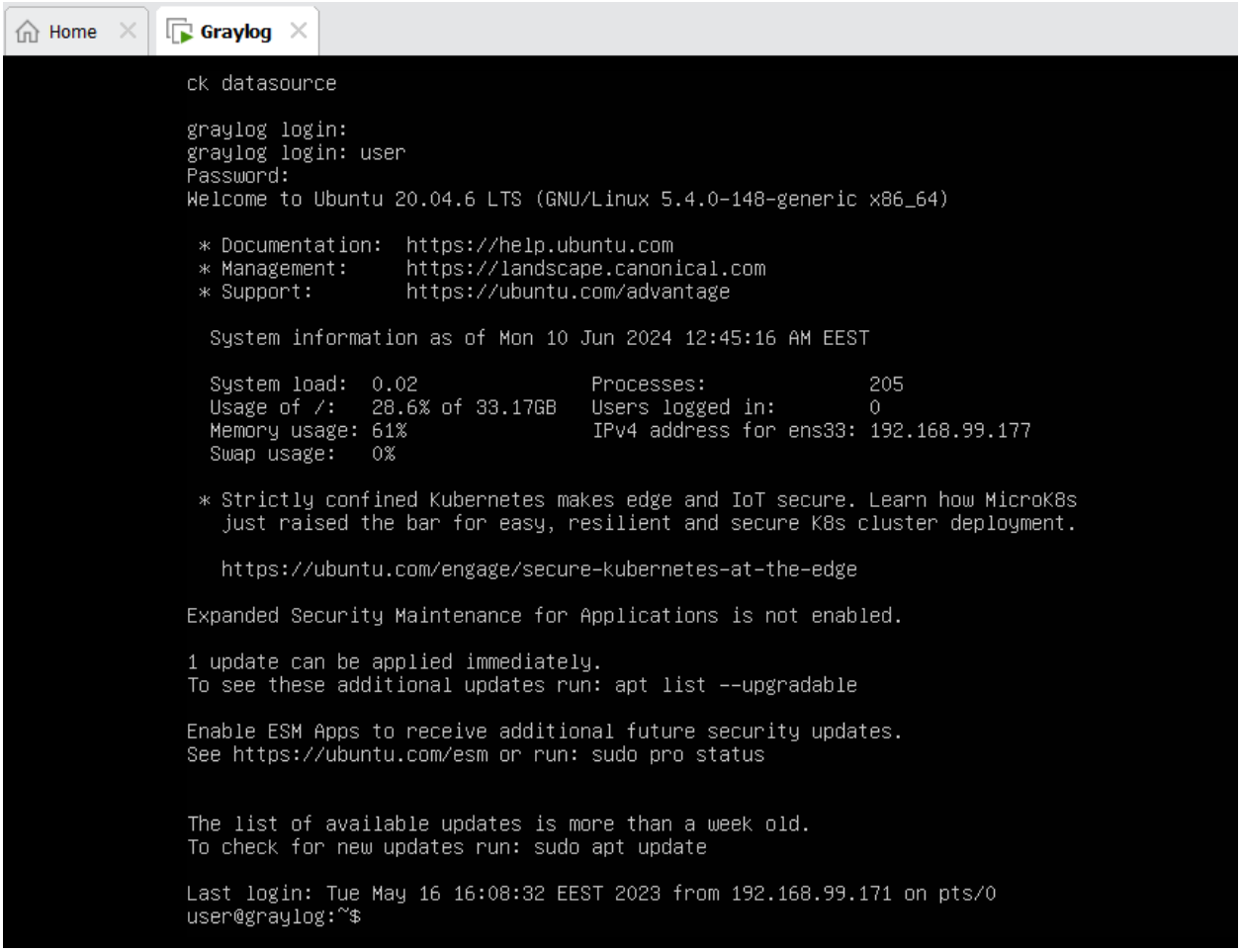


Figure 4.7 – Logging in to the server operating system

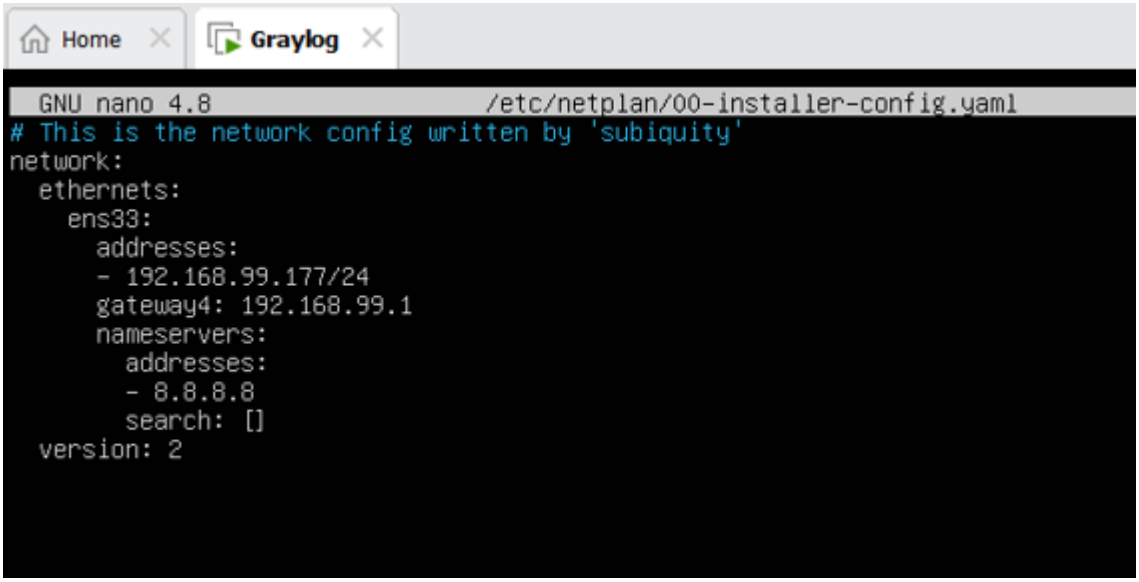


Figure 4.8 – Opening the /etc/netplan/00-installer-config.yaml file

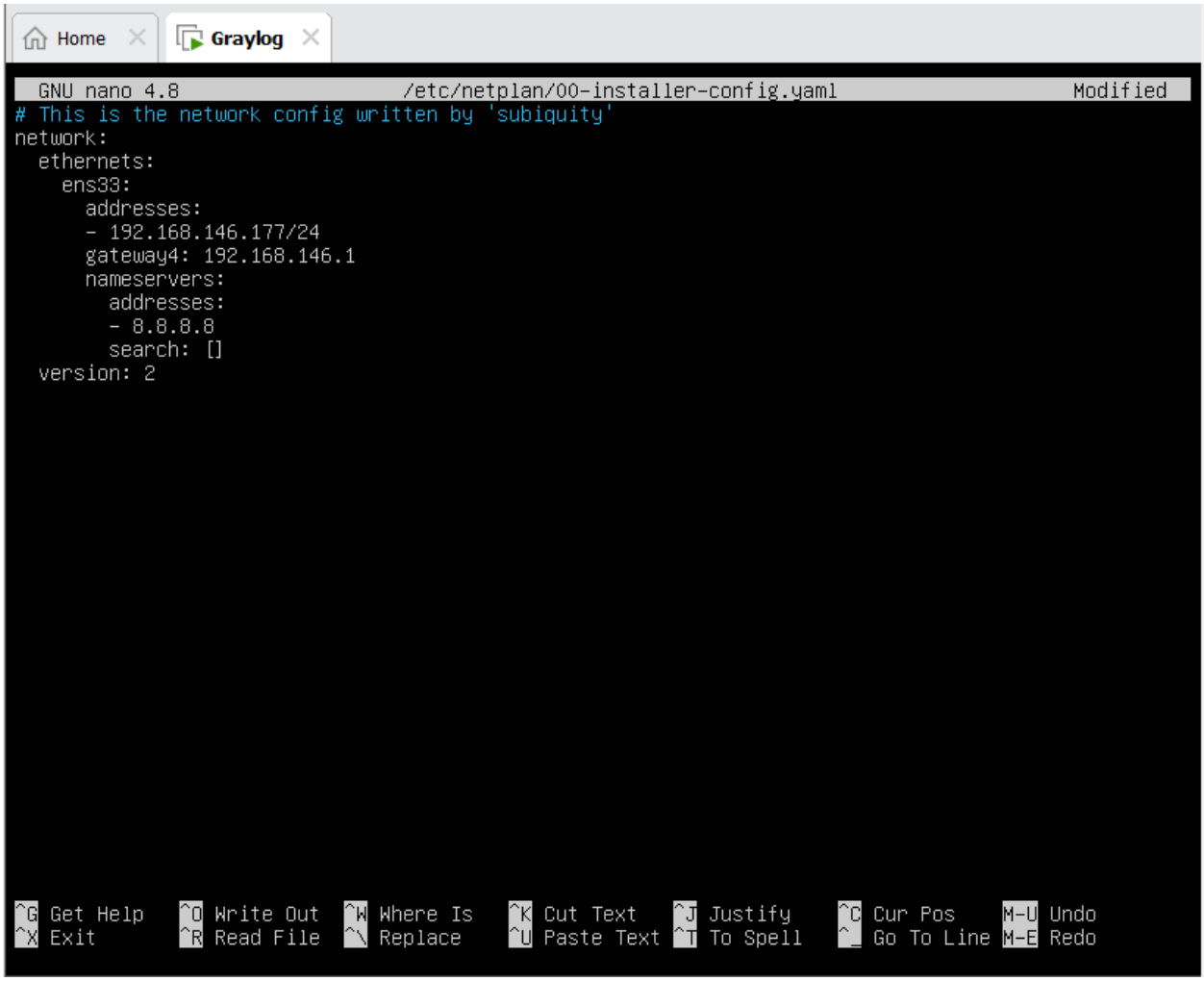


Figure 4.9 – Changing the IP address to the previously selected address, 192.168.146.177

```
root@graylog:/home/user#
root@graylog:/home/user# netplan apply
root@graylog:/home/user# nano /etc/graylog/server/server.conf
```

Figure 4.10 – Opening the /etc/graylog/server/server.conf file

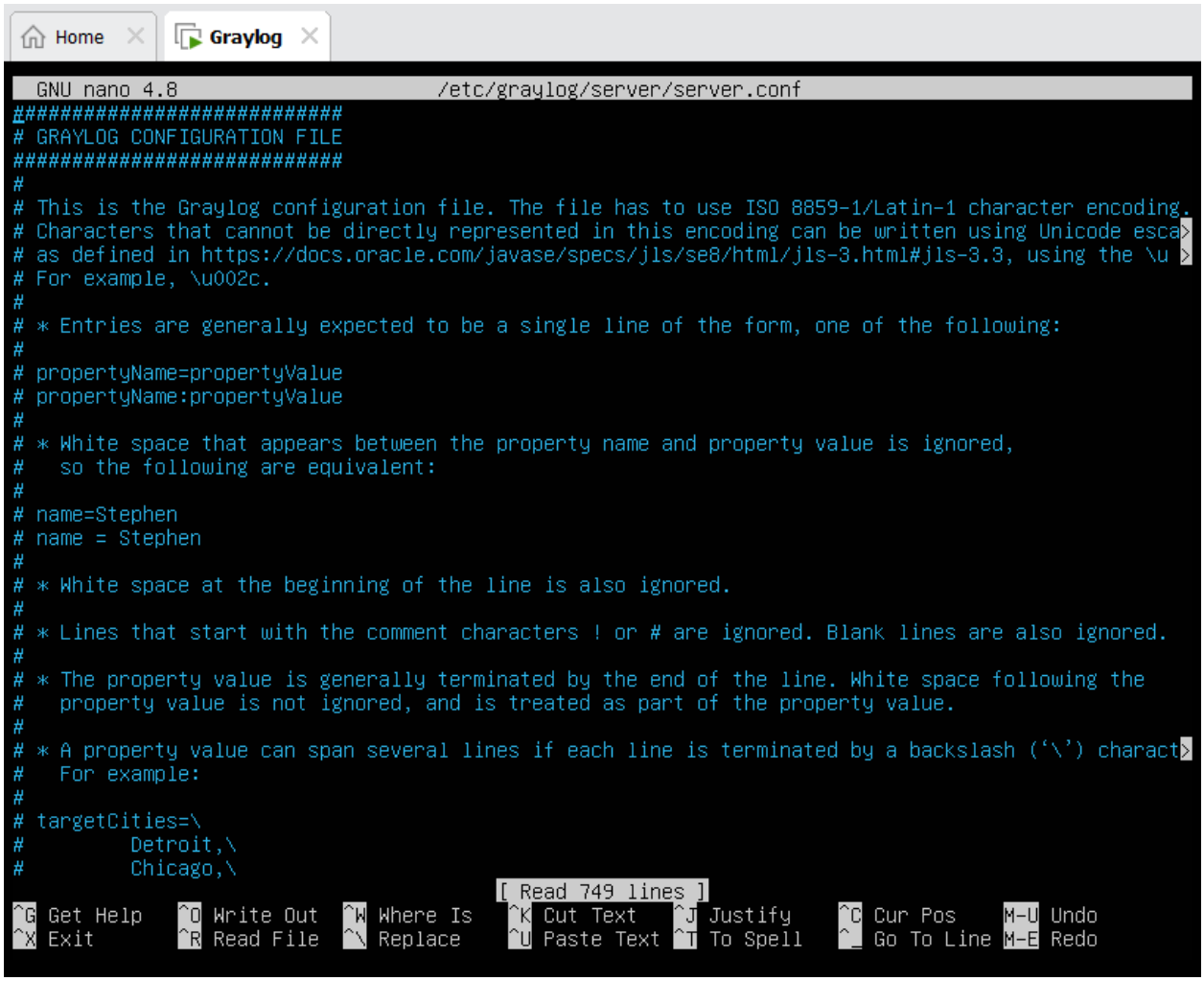


Figure 4.11 – Opening the /etc/graylog/server/server.conf file

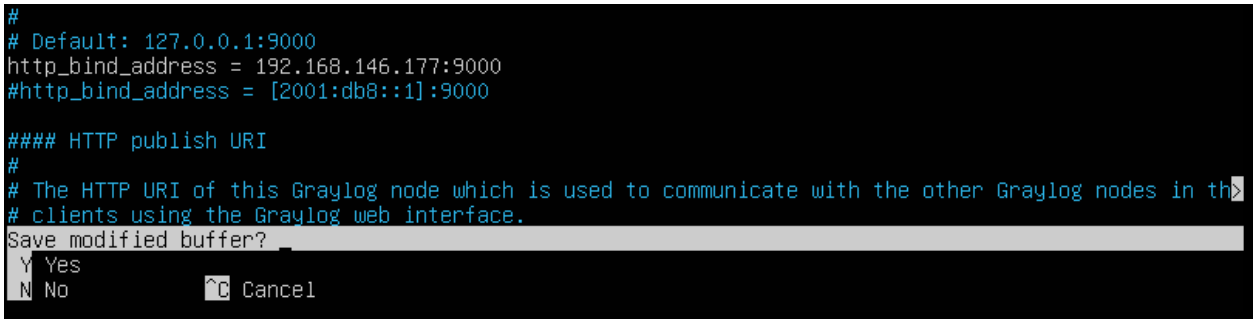


Figure 4.12 – Changing the value of http_bind_address to the previously selected address, 192.168.146.177, and saving the changes



4.13 – Restarting the virtual machine

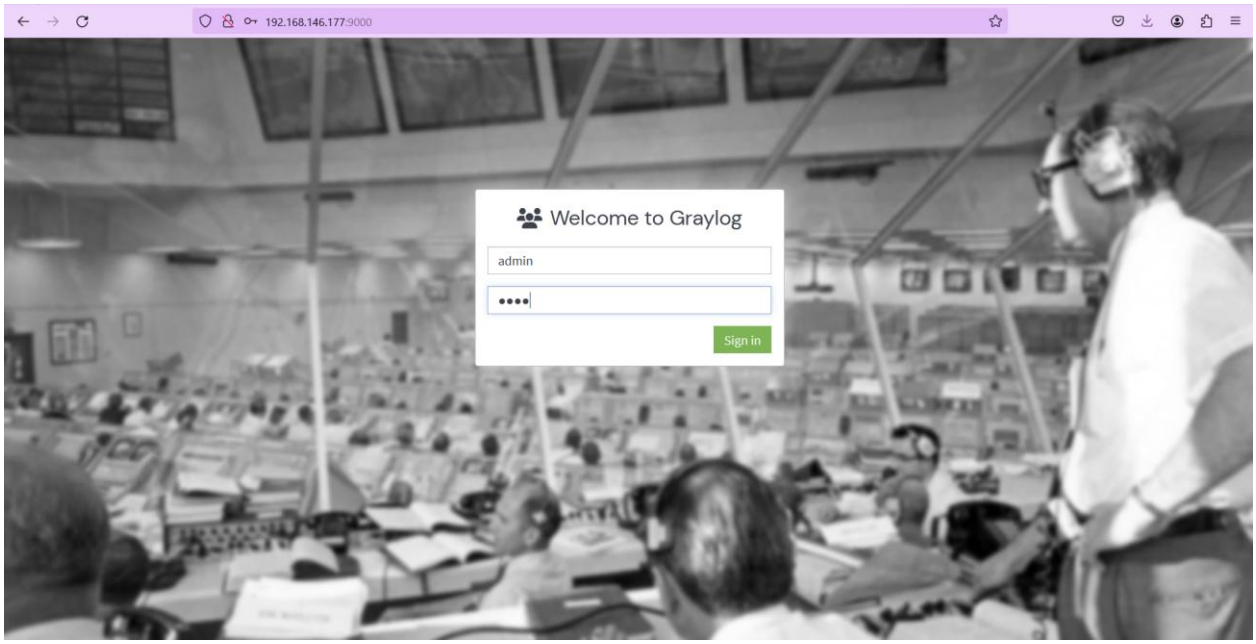


Figure 4.14 – Accessing the web interface at http://192.168.146.177:9000

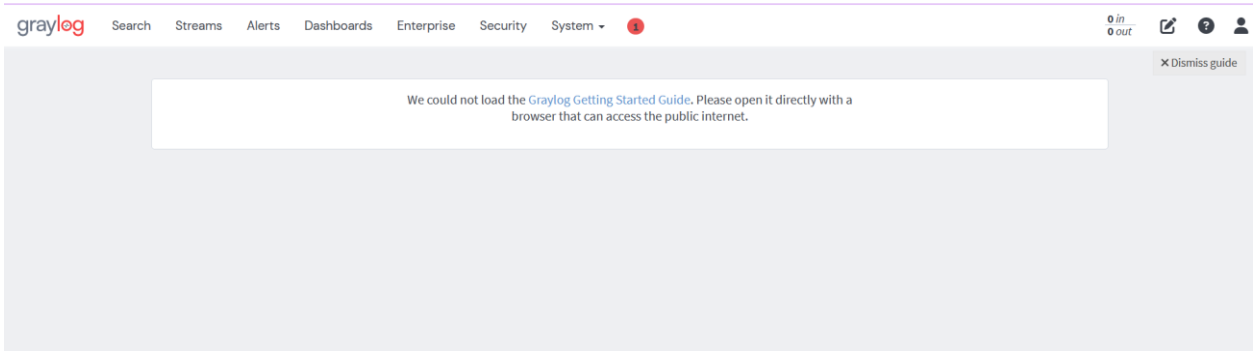


Figure 4.15 – Main page

After configuring the server and changing its IP address, the web interface was successfully opened in a browser using the server's IP address.

4.2.3 Creating an input for log data

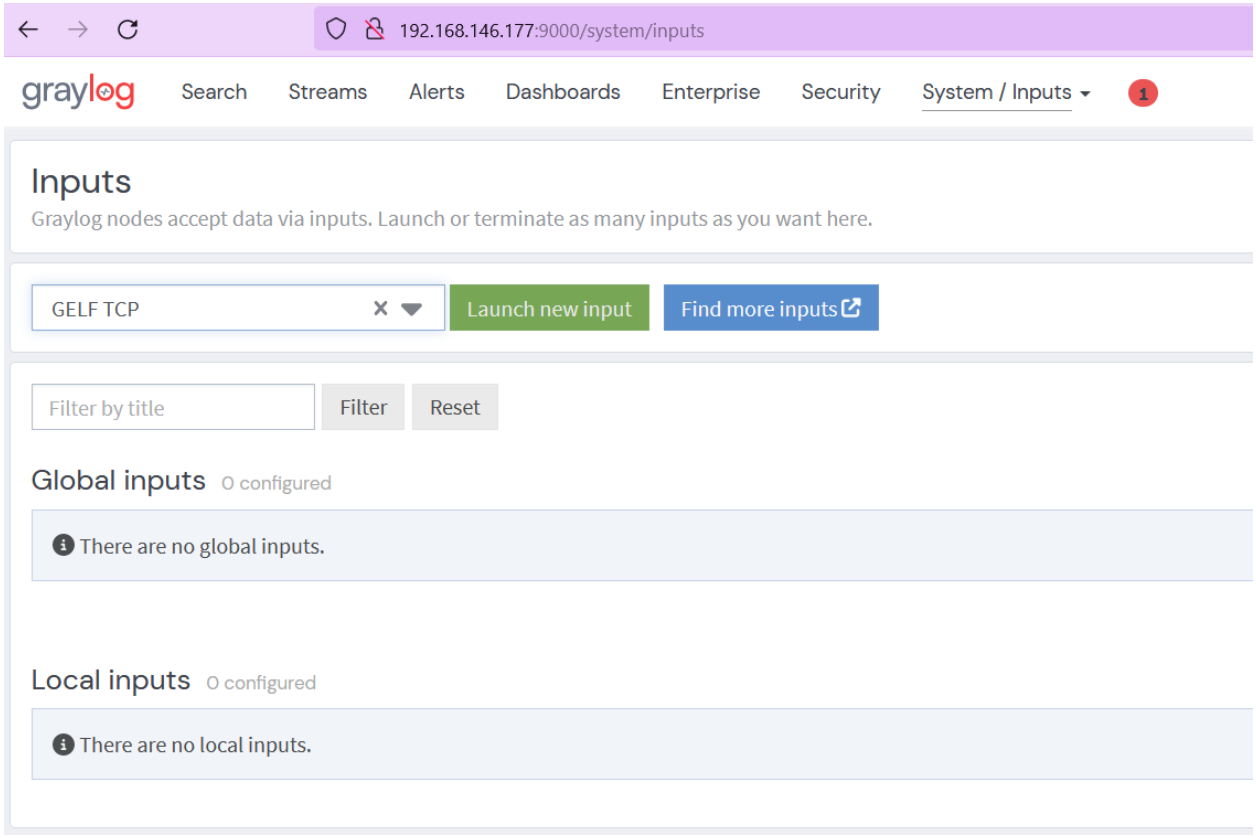


Figure 4.16 – Creating an input and selecting its type

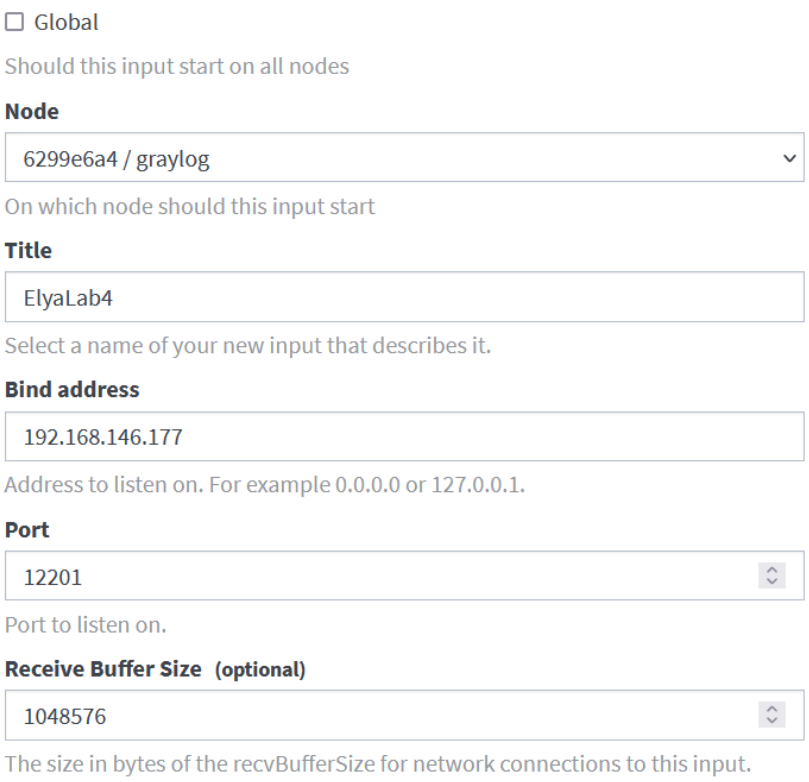


Figure 4.17 – Configuring the input parameters

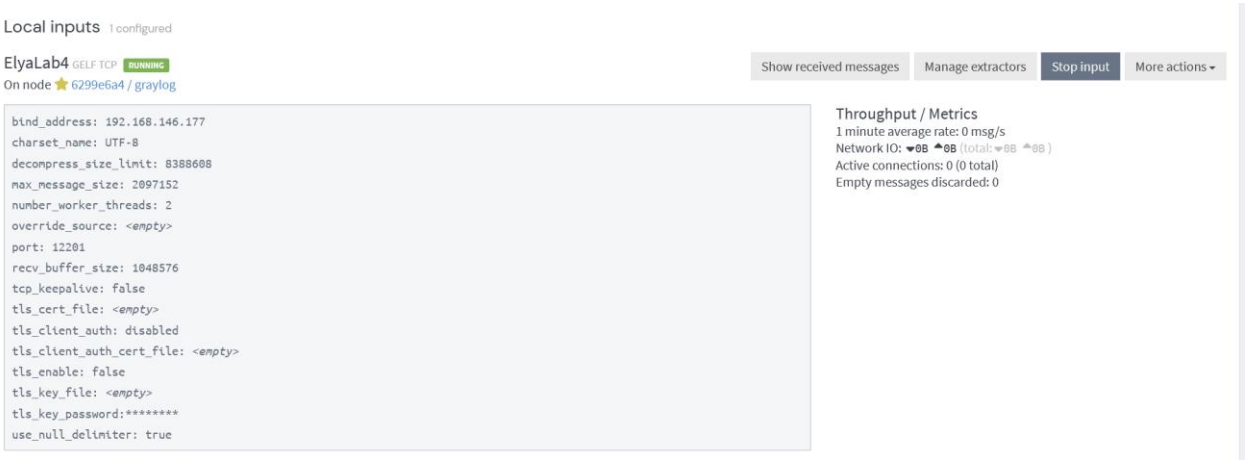
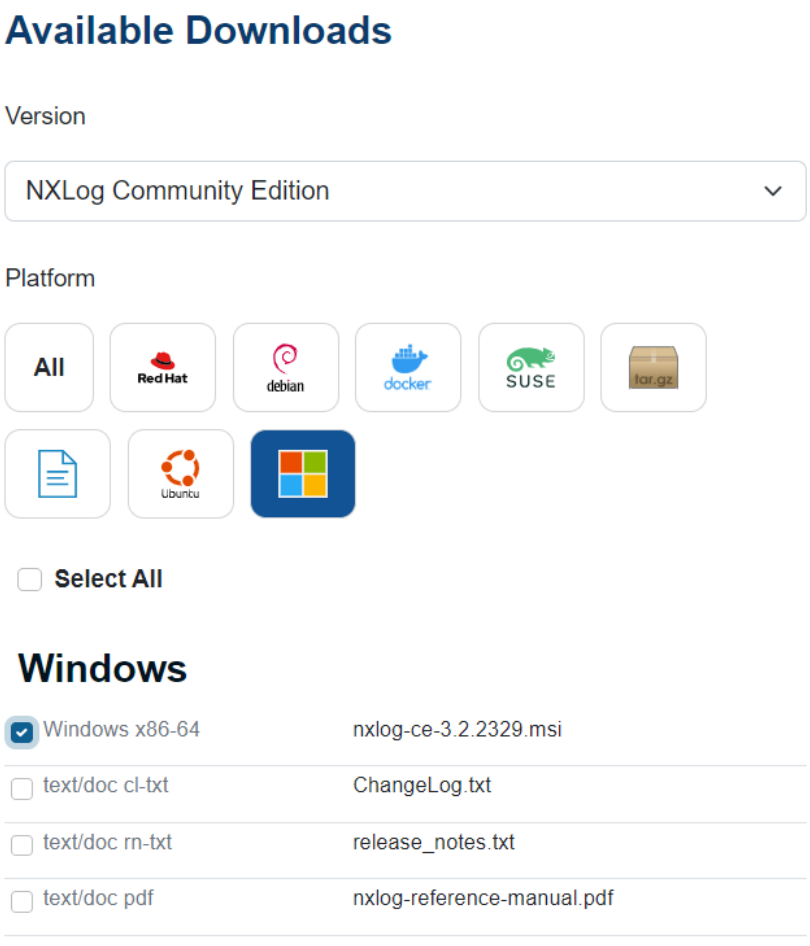


Figure 4.18 – Result of creating the input

4.2.4 Configuring the Windows PC



NXLog Figure 4.19 – Downloading NXLog

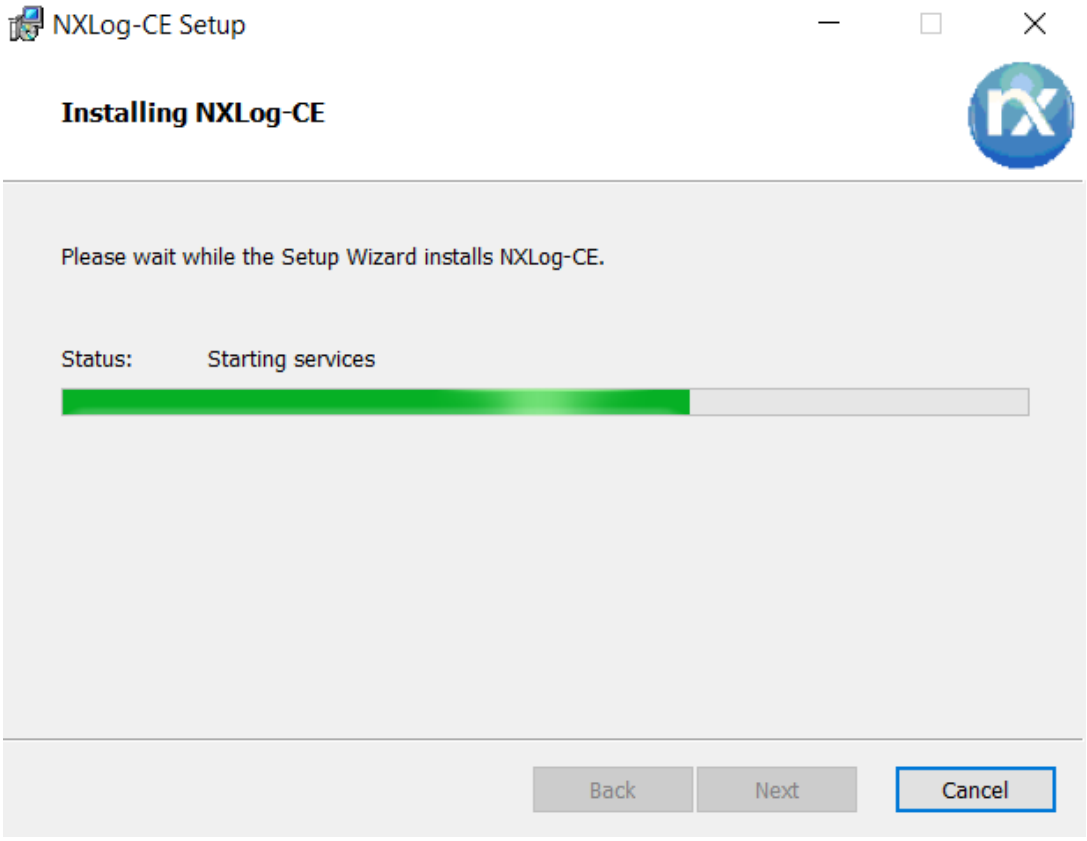


Figure 4.20 – Installing NXLog

4.2.5 Configuring NXLog

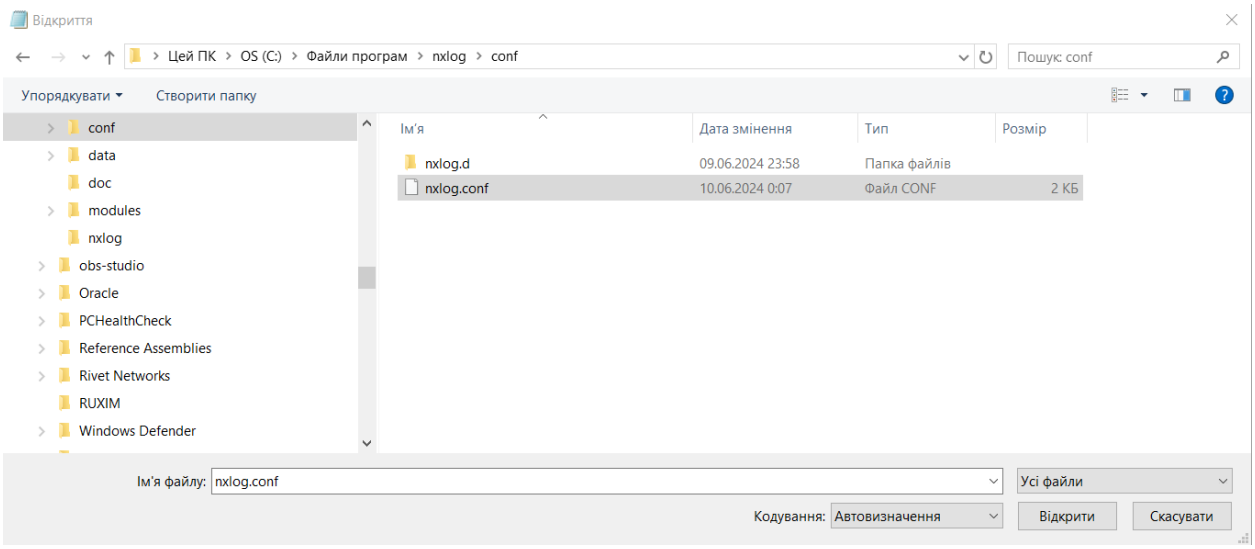


Figure 4.21 – Opening the nxlog.conf file in Notepad

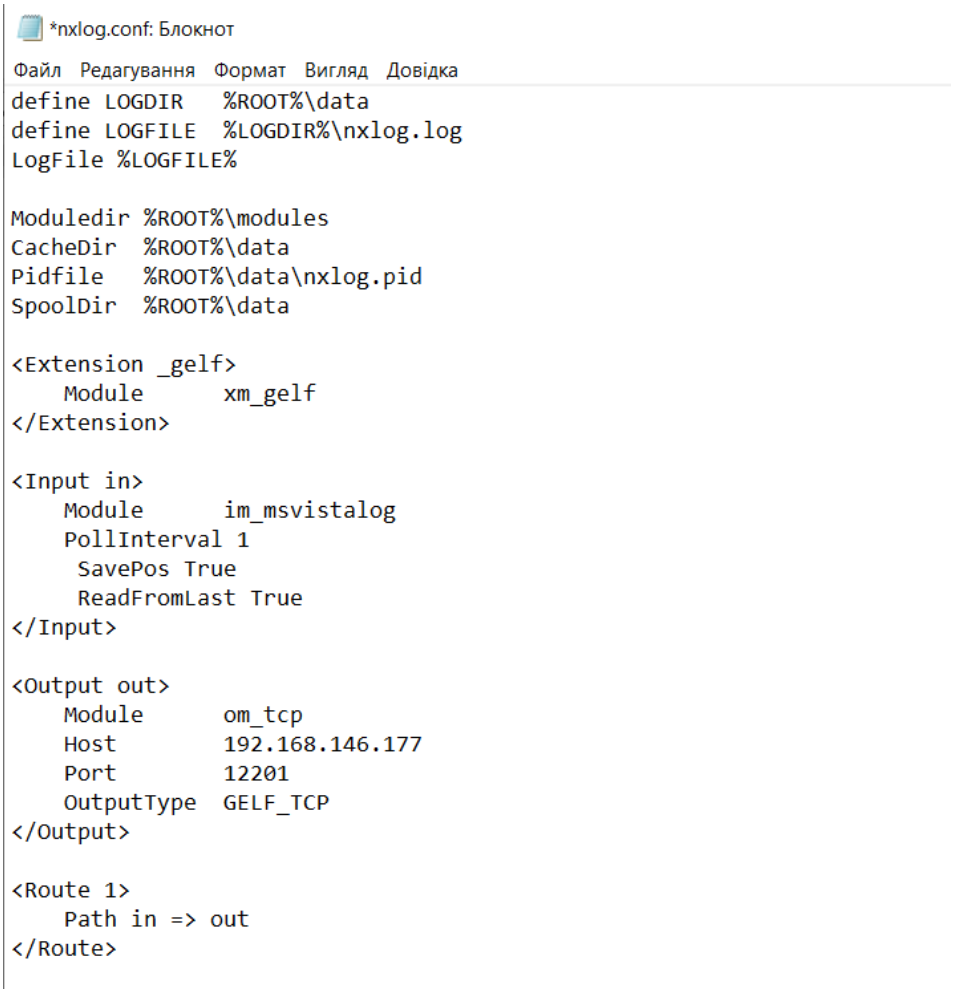


Figure 4.22 – Changes made to the nxlog.conf file

The configuration defines the directories used by NXLog and the location of its own log file. The xm_gelf module enables GELF formatting, while im_msvistalog collects Windows event logs. The om_tcp module sends the logs to the Graylog server over TCP on port 12201. The final section connects the input and output, allowing NXLog to forward Windows logs to Graylog.

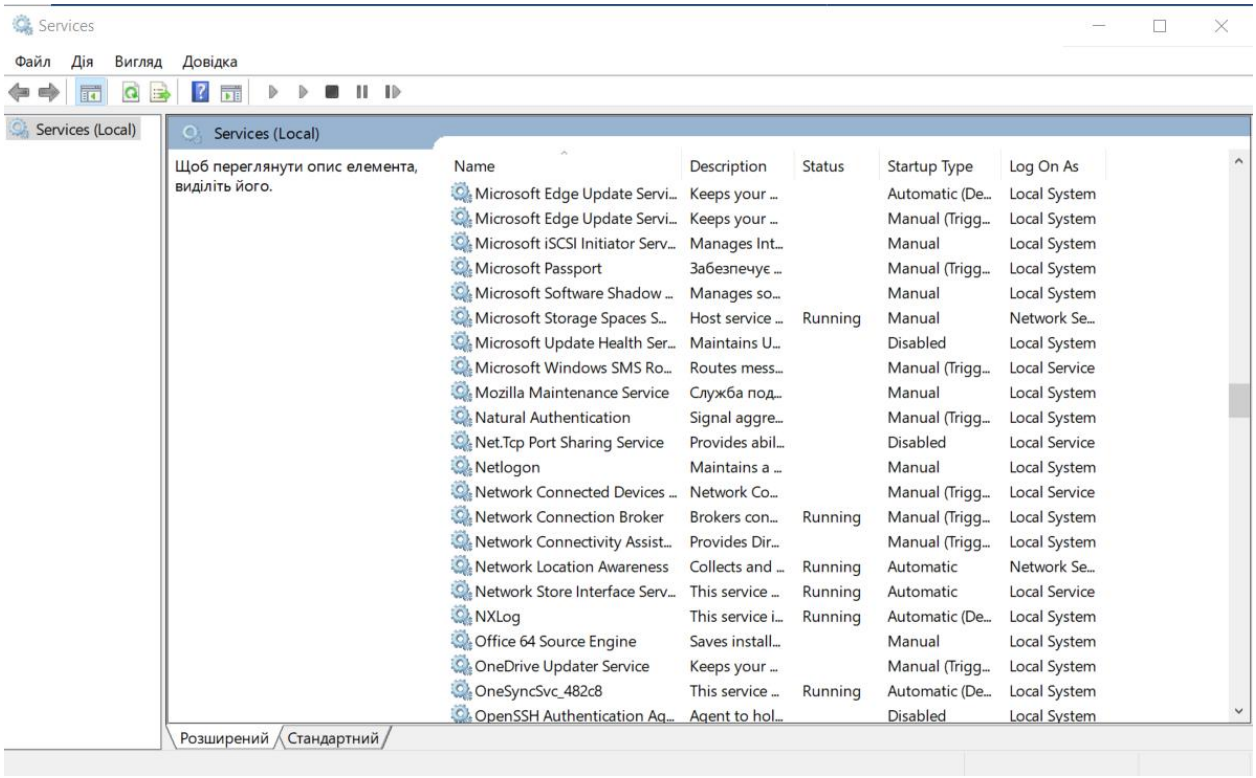


Figure 4.23 – Opening the Windows Services management console with administrator privileges

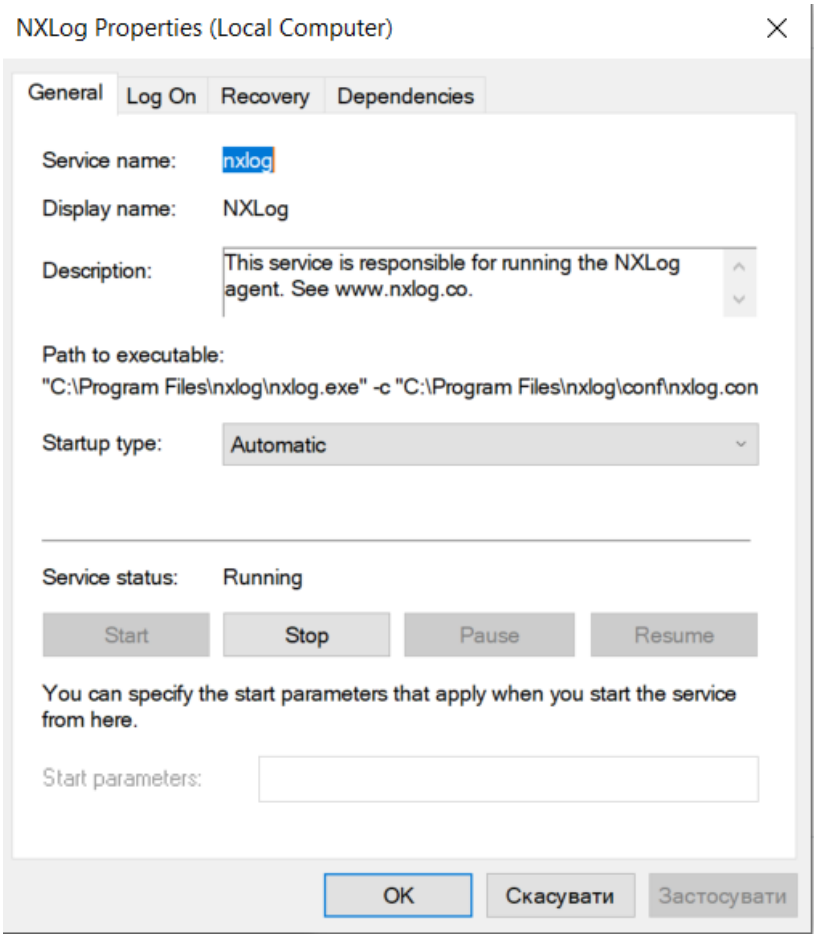


Figure 4.24 – Changing the properties of the NXLog service

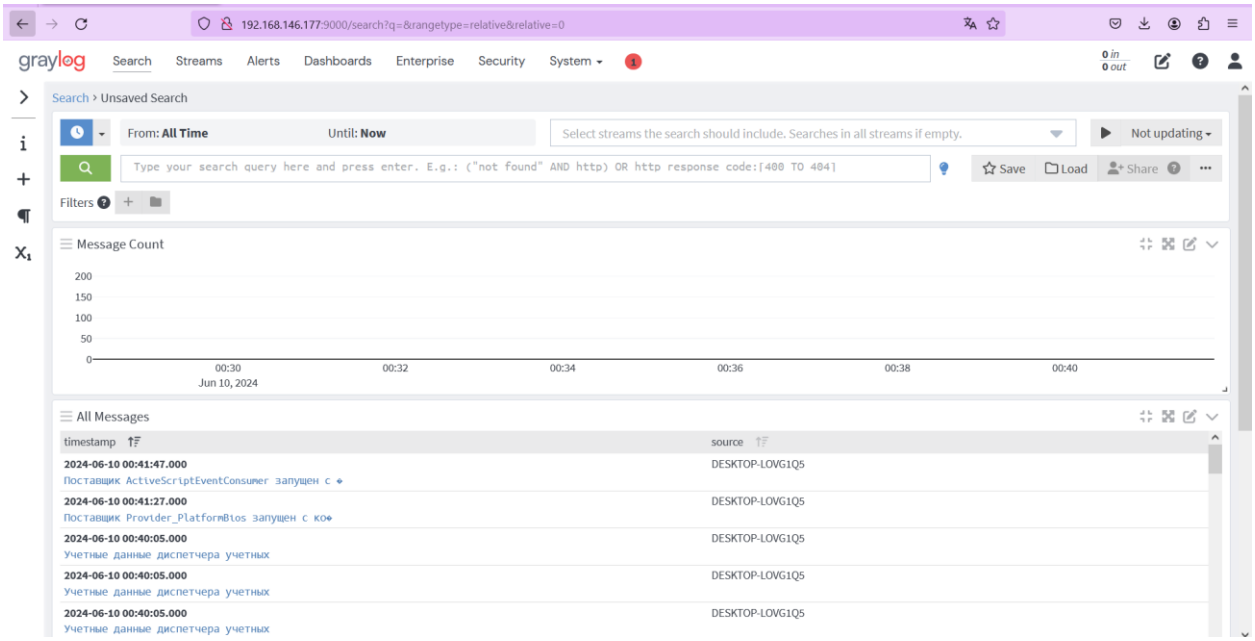


Figure 4.29 – Logs appearing in the “Search” section

After configuring NXLog and starting the service, log data began to be forwarded to the Graylog server. The logs appeared in the “Search” section, indicating that the log collection and forwarding configuration was working successfully.

4.2.6 Independent tasks

The task was to generate events in Windows and monitor them in the SIEM system. The details of each event were recorded in the report.

The following events were monitored:

1. Event ID 4720 – User account creation

2. Event ID 4723 – Attempt to change a user's password

3. Event ID 4625 – Failed logon attempt

4. Event ID 4726 – User account deletion

4.2.6.1 User account creation (event ID 4720)

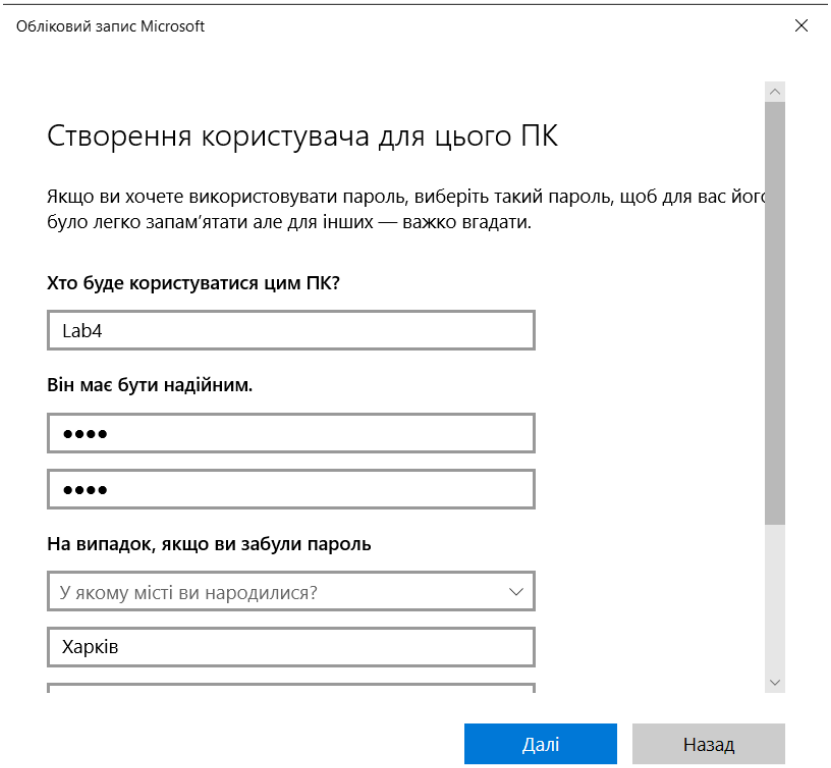


Figure 4.25 – Creating a new user account

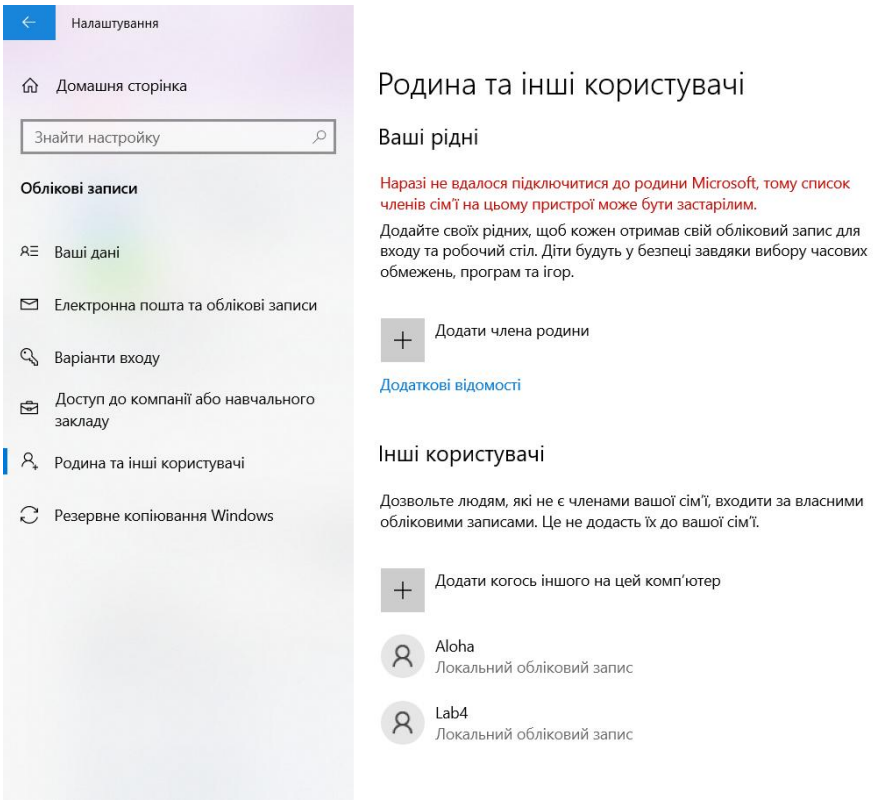


Figure 4.26 – Result of creating a new user account

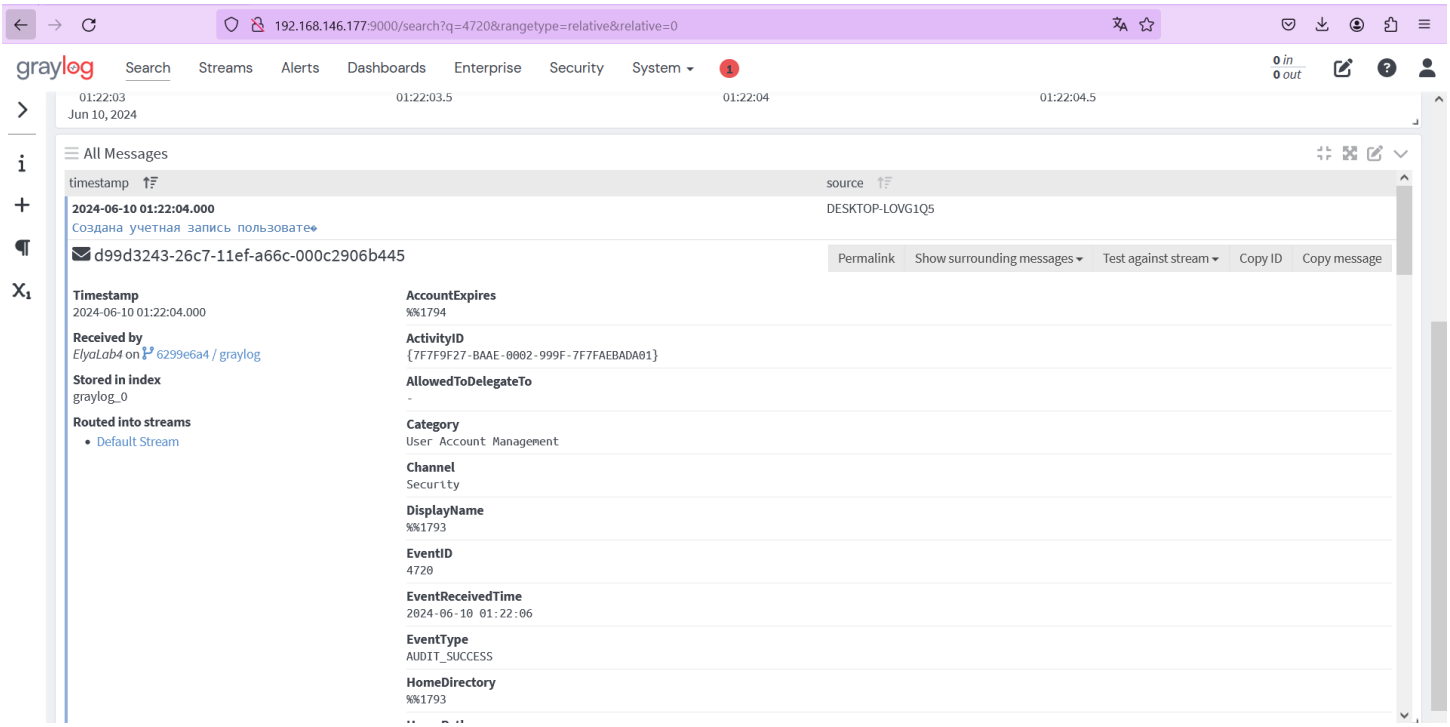


Figure 4.27 – Viewing event ID 4720 generated when a user account is created

4.2.6.2 Changing a user's password (event ID 4723)

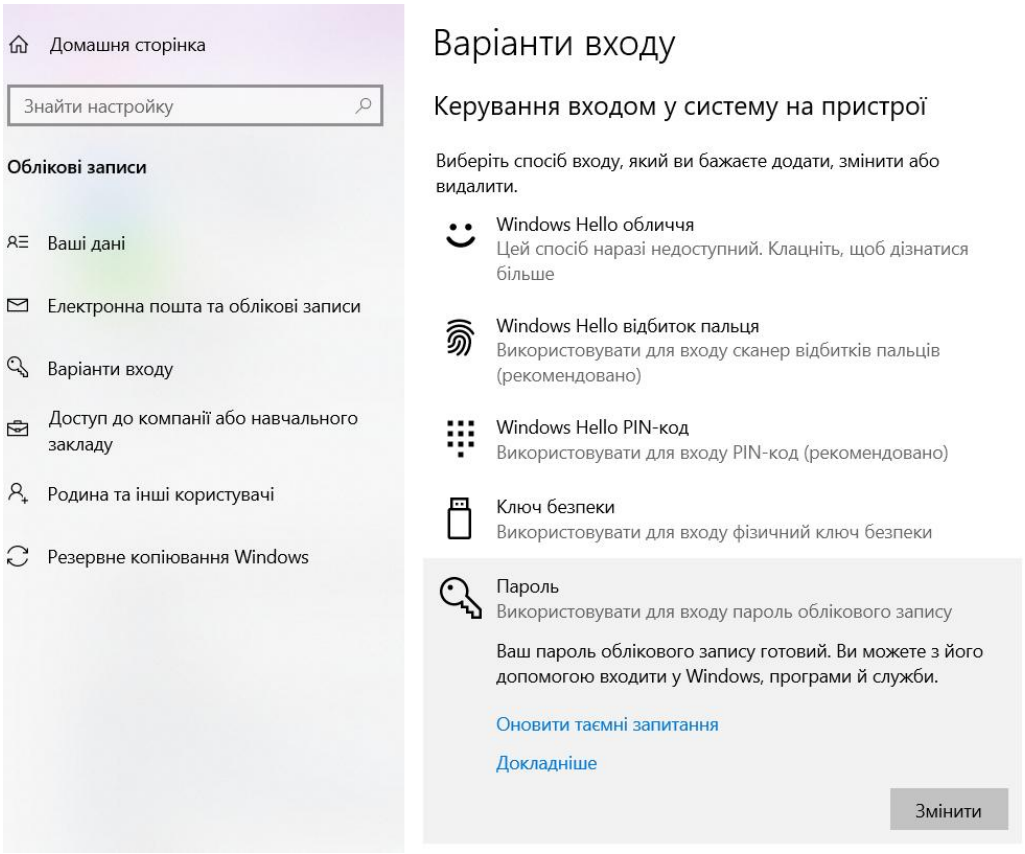


Figure 4.28 – Changing a user's password

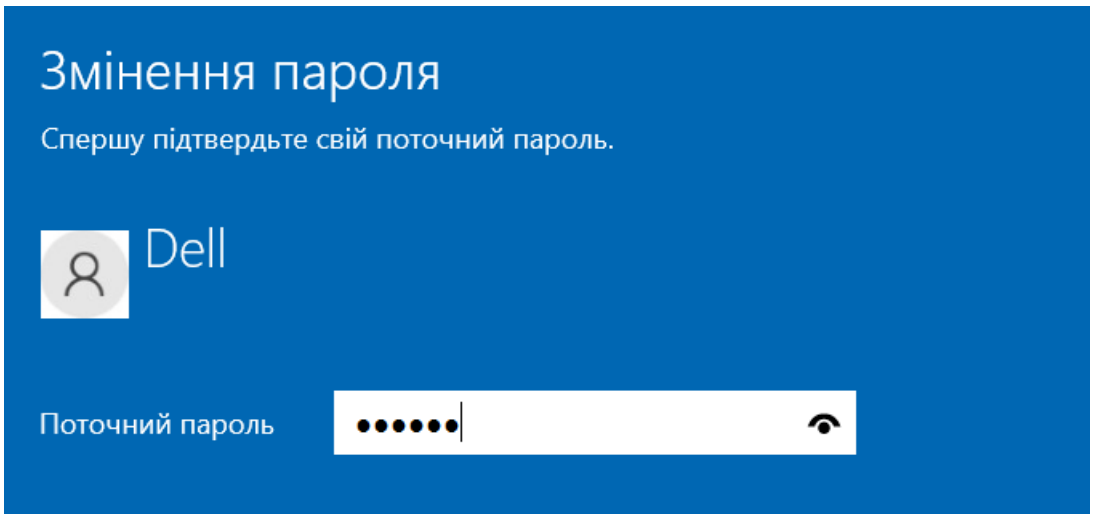


Figure 4.29 – Entering the current password

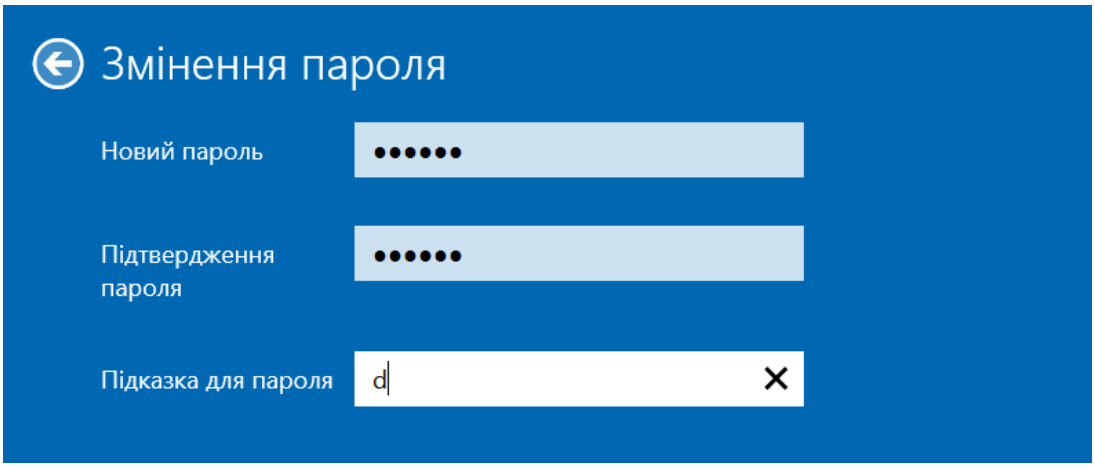


Figure 4.30 – Entering the new password

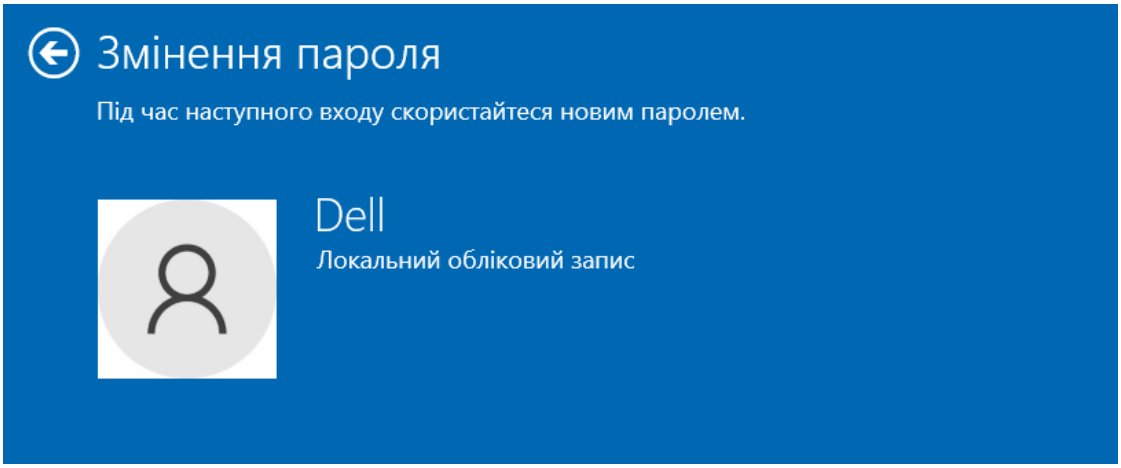


Figure 4.31 – Confirming the password change

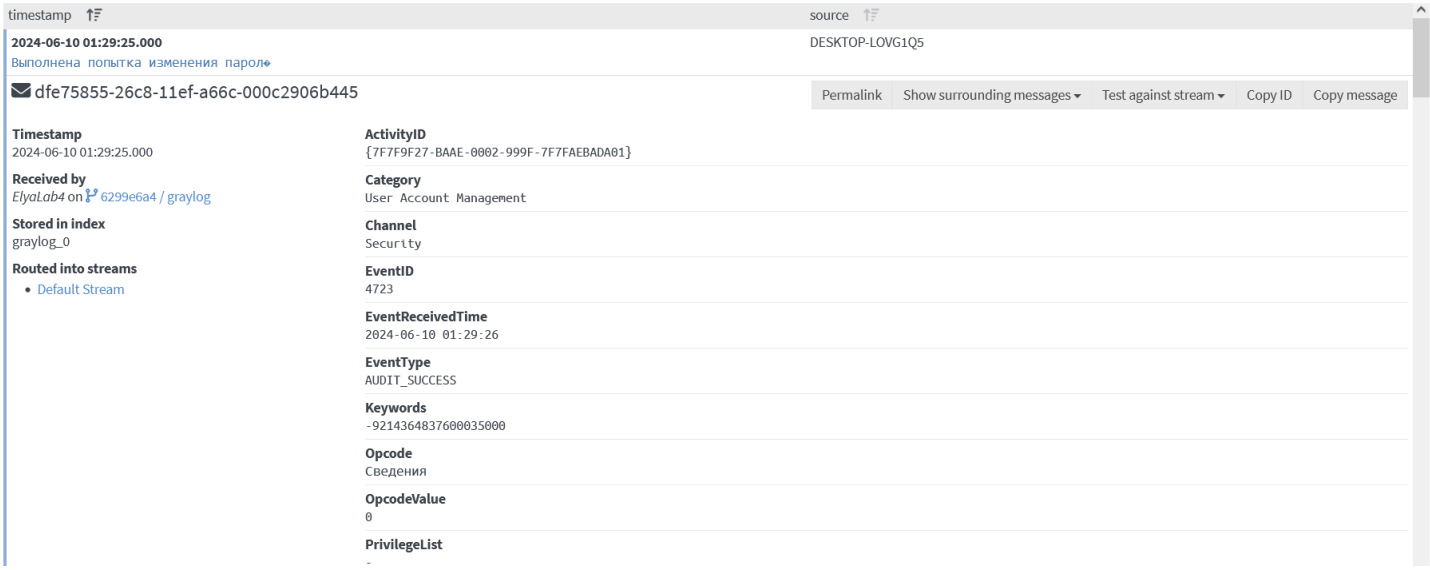


Figure 4.32 – Viewing event ID 4723 generated during a password change attempt

4.2.6.3 Failed authentication (event ID 4625)

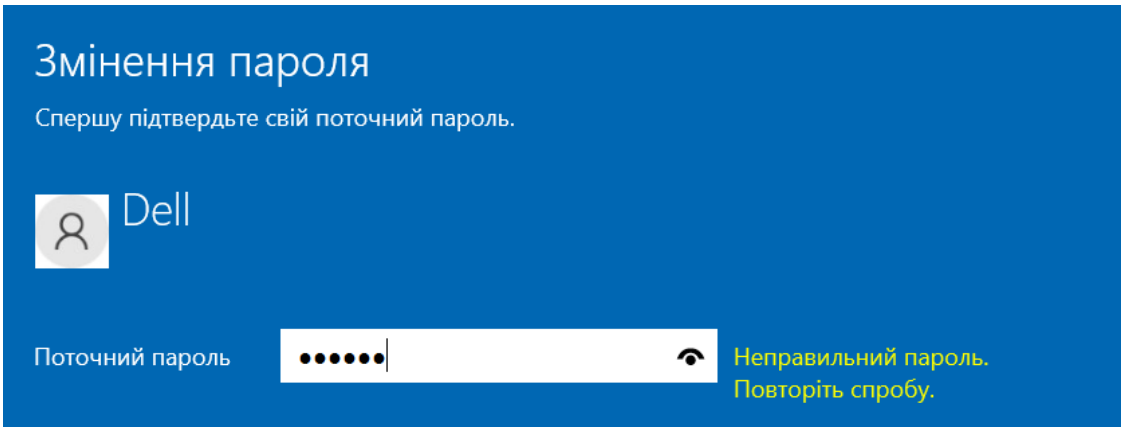


Figure 4.33 – Entering an incorrect password when attempting to change a password

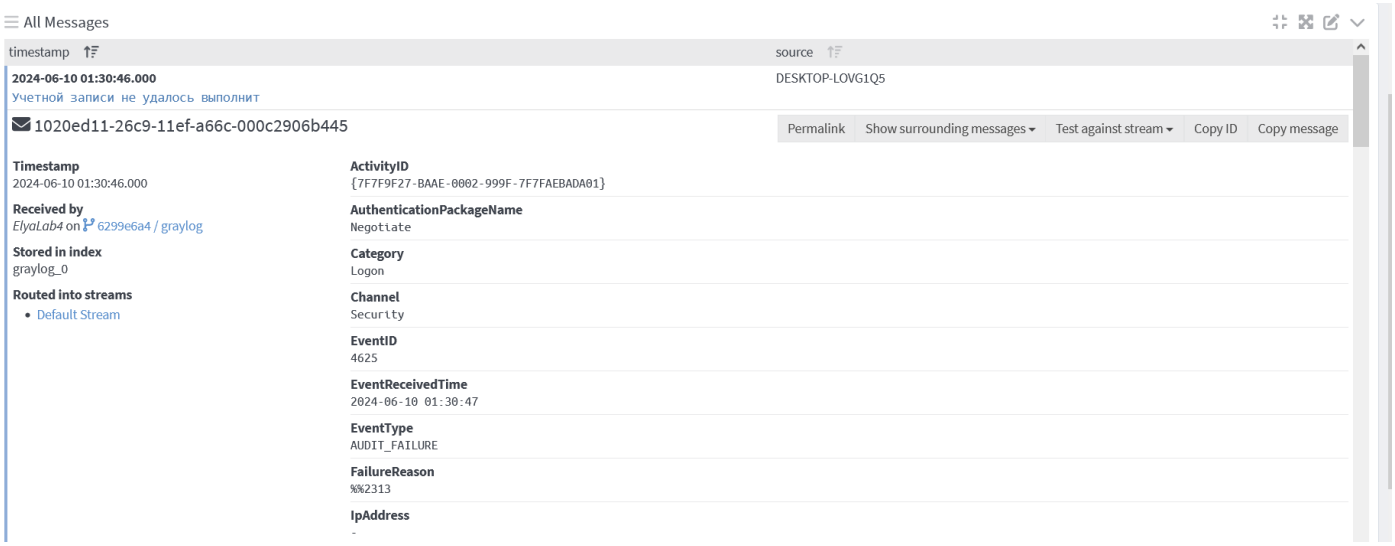


Figure 4.34 – Viewing event ID 4625 generated after a failed authentication attempt

4.2.6.4 User account deletion (event ID 4726)

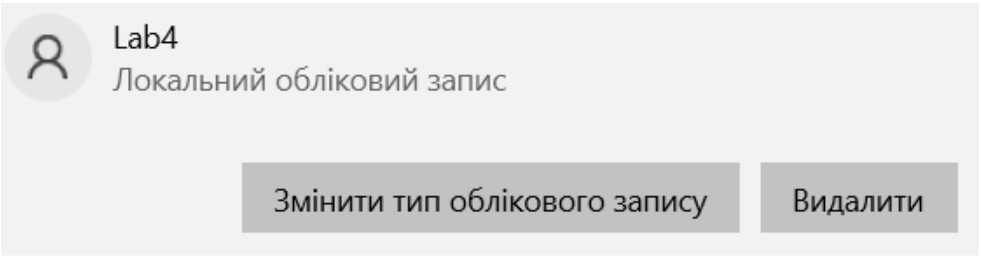


Figure 4.35 – Deleting a user account

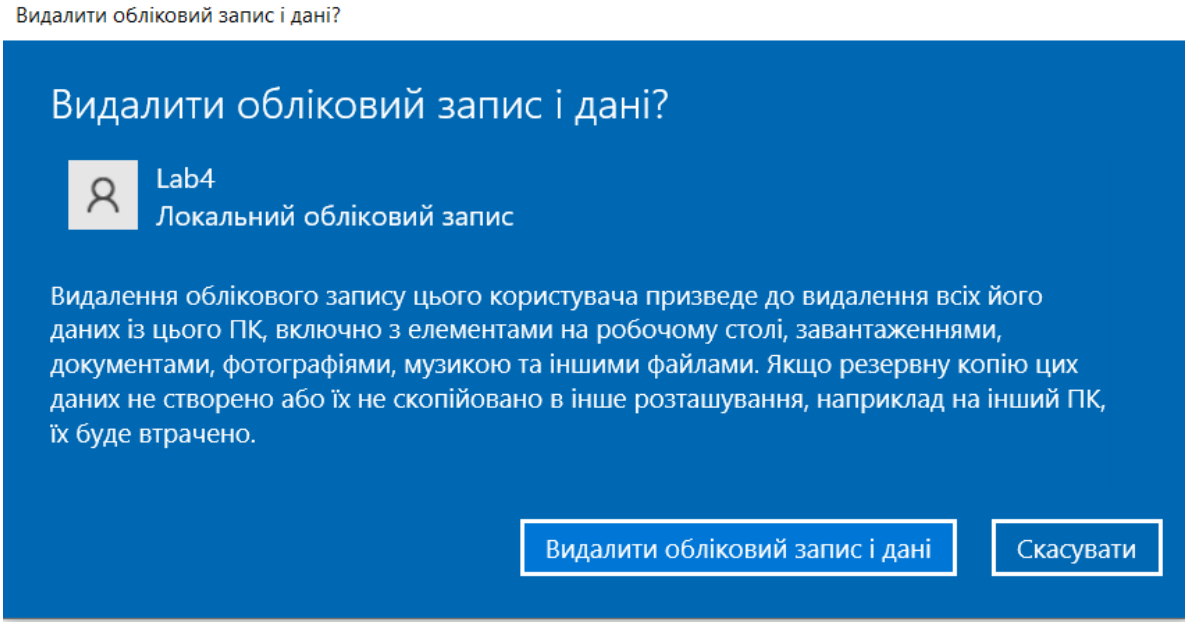


Figure 4.36– Confirming the deletion of a user account

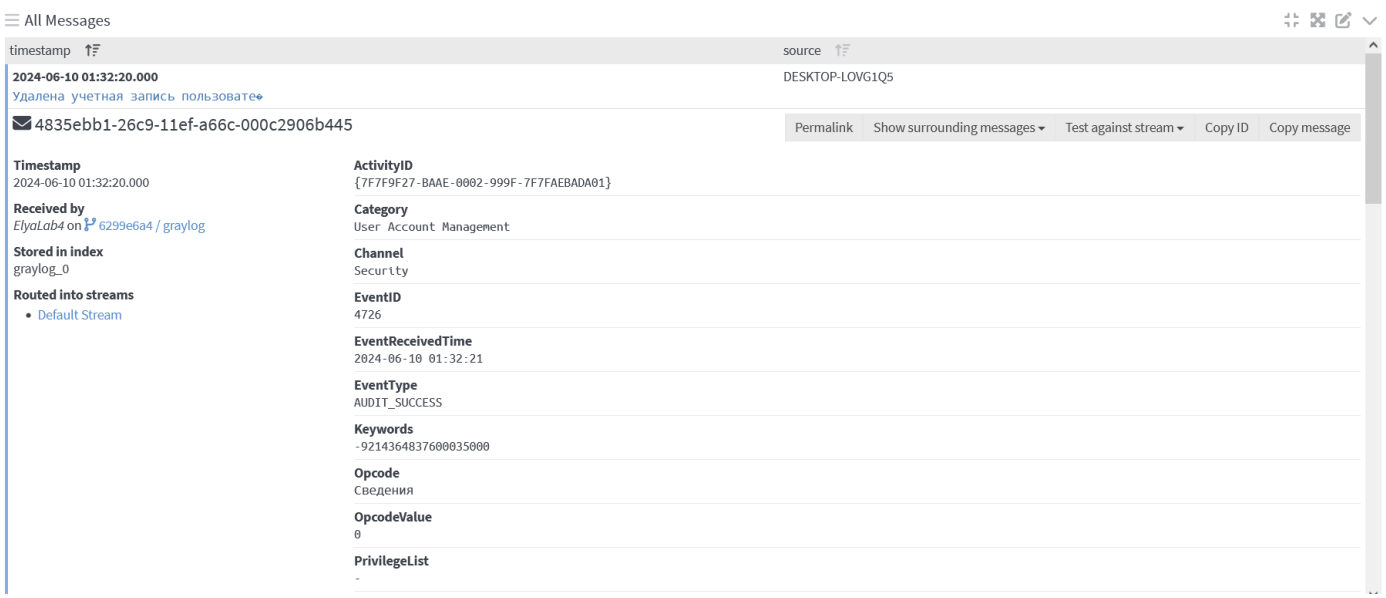


Figure 4.37 – Viewing event ID 4726 generated when a user account is deleted

By creating, modifying, and deleting a user account and entering an incorrect password, I generated events that could be monitored in the “Search” section of Graylog.

4.3 Conclusion

During this laboratory, I learned how to use a SIEM system to monitor and detect security-related events. I installed and configured Graylog in a virtual machine, set up log collection, and configured NXLog to forward Windows event logs to the server. Finally, I generated four types of events and found them in Graylog, gaining practical experience in log monitoring and event analysis.